

A_sub commutator closures steps 3-5 — $\hat{\gamma}^5$ with $\hat{T}$, $\hat{C}$, $\hat{P}_{op}$: substrate-Pin(2)-grading vs discrete symmetries

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2026-05-26 Tuesday EDT (~08:55 EDT via date)

A_sub commutator closures steps 3-5 — $\hat{\gamma}^5$ vs $\hat{T}$, $\hat{C}$, $\hat{P}_{op}$

1. Setup — $\hat{\gamma}^5$ and discrete symmetries

1.1 Pin(2) Z_2 chirality $\hat{\gamma}^5$ (T2471 RATIFIED)

$\hat{\gamma}^5 V_{(m_1, m_2)} = \varepsilon_K \cdot V_{(m_1, m_2)}$, $\varepsilon_K = +1$ (integer K-type, boson), -1 (half-integer K-type, fermion)

$\hat{\gamma}^5$ is involutive: $(\hat{\gamma}^5)^2 = 1$. Hermitian. Self-adjoint.

v0.7 disambiguation flag (added 2026-05-26 PM per Cal #29 first formal application catch): the ε_K identifying “+1 boson / -1 fermion at K-type sublattice level” is σ_{BF} (Pin(2) Z_2 sublattice grading), NOT γ^5 as Dirac chirality. These are DISTINCT operators:

- σ_{BF} (Pin(2) Z_2 grading): integer-vs-half-integer K-type sublattice partition; bose/fermi distinguisher; commutes with T, C, P (preserves particle type)
- γ^5 (Dirac chirality): L/R Weyl spinor partition WITHIN fermion sublattice; undefined on bosons; anti-commutes with T, C (standard Dirac algebra)

Cal #132 verified the algebraic relations in Sections 2-5 below for γ^5 as Dirac chirality. The substantive content stands at SVC for that interpretation. The K-type sublattice identification ($\varepsilon_K = \pm 1$ for boson/fermion) uses σ_{BF} ; γ^5 -as-Dirac-chirality is the operator whose anti-commutation relations with T, C and commutation with P are computed below.

In v0.8+ formal cleanup: σ_{BF} and γ^5 will be explicitly separate notations throughout A_sub. Steps 3-5 results below remain SVC at γ^5 -as-Dirac-chirality interpretation.

1.2 Time-reversal $\hat{T}$ (T2433 STRUCTURALLY VERIFIED)

$\hat{T}$ is anti-unitary (anti-linear, $\hat{T}(a \cdot |\psi\rangle) = a^* \cdot \hat{T}|\psi\rangle$). On Wallach K-types:

$$\hat{T} V_{(m_1, m_2)} = (\text{Pin}(2)\text{-}\hat{T}\text{-phase}) \cdot \bar{V}_{(m_1, -m_2)}$$

where $\bar{V}_{(m_1, -m_2)}$ is the complex-conjugate-representation K-type with m_2 sign-flipped (T flips angular-momentum-like quantities; on $SO(2)$, $\hat{L} \rightarrow -\hat{L}$ which negates m_2).

The $\text{Pin}(2)\text{-}\hat{T}\text{-phase}$ depends on the Z_2 grading: $\hat{T}$ flips the orientation of S^1 rotations, hence flips the $\text{Pin}(2)$ Z_2 grading on fermion K-types.

1.3 Charge-conjugation $\hat{C}$ (T2434 STRUCTURALLY VERIFIED)

$\hat{C}$ is anti-linear (in standard convention) and takes particle $\leftrightarrow$ antiparticle:

$$\hat{C} V_{(m_1, m_2)} = (\text{Pin}(2)\text{-}\hat{C}\text{-phase}) \cdot V_{(m_1, -m_2)}$$

$\hat{C}$ flips charge $q \rightarrow -q$ (i.e., $m_2 \rightarrow -m_2$ on Wallach K-types) AND flips chirality (left-handed particle is C-conjugate to right-handed antiparticle in standard Dirac analysis).

1.4 Substrate-parity $\hat{P}_{\text{op}}$ (T2472 STRUCTURALLY VERIFIED)

$$\hat{P}_{\text{op}} = \hat{\gamma}^5 \circ \sigma$$

where σ is the Möbius involution on D_{IV}^5 's $SO(2)$ factor ($z \rightarrow -\bar{z}$), and $\hat{\gamma}^5$ is the $\text{Pin}(2)$ Z_2 lift. Per step 1 $[\hat{Q}, \hat{P}_{\text{op}}]$ closure:

$$\hat{P}_{\text{op}} V_{(m_1, m_2)} = \varepsilon_K \cdot (-1)^{m_2} \cdot V_{(m_1, -m_2)}$$

2. Step 3 — $[\hat{\gamma}^5, \hat{T}]$

2.1 Computation

$$\text{Apply } \hat{\gamma}^5 \circ \hat{T}: \hat{\gamma}^5 \hat{T} V_K = \hat{\gamma}^5 \cdot (\text{Pin}(2)\text{-}\hat{T}\text{-phase}) \cdot \bar{V}_{(m_1, -m_2)} = (\text{Pin}(2)\text{-}\hat{T}\text{-phase}) \cdot \varepsilon_{\{\sigma K\}} \cdot \bar{V}_{(m_1, -m_2)}$$

where $\varepsilon_{\{\sigma K\}}$ is the $\text{Pin}(2)$ Z_2 grading of the σ -flipped K-type. The key question is whether $\varepsilon_{\{\sigma K\}} = \varepsilon_K$ or $\varepsilon_{\{\sigma K\}} = -\varepsilon_K$.

$\hat{T}$ flips the orientation of S^1 rotations (T-invariant of $\hat{L}$: $\hat{L} \rightarrow -\hat{L}$). The $\text{Pin}(2)$ Z_2 grading ε measures the relative orientation between two $\text{Pin}(2)$ elements; under T flipping the rotation generator, the grading sign FLIPS:

$$\varepsilon_{\{\sigma K\}} = -\varepsilon_K \text{ under T-action}$$

(This is the substrate-mechanism reading; consistent with standard Dirac matrix algebra where γ^5 anti-commutes with γ^0 and time-reversal involves $\gamma^0 \circ K$.)

$$\text{Apply } \hat{T} \circ \hat{\gamma}^5 \text{ (with } \hat{T} \text{ anti-linear flipping the sign of } i\text{): } \hat{T} \hat{\gamma}^5 V_K = \hat{T}(\varepsilon_K \cdot V_K) = \varepsilon_K \cdot \hat{T} V_K \text{ (since } \varepsilon_K = \pm 1 \text{ is real)} = \varepsilon_K \cdot (\text{Pin}(2)\text{-}\hat{T}\text{-phase}) \cdot \bar{V}_{(m_1, -m_2)}$$

Anti-commutator: $\{\hat{\gamma}^5, \hat{T}\} V_K = \hat{\gamma}^5 \hat{T} V_K + \hat{T} \hat{\gamma}^5 V_K = (\text{Pin}(2)\text{-}\hat{T}\text{-phase}) \cdot \bar{V}_{(m_1, -m_2)} \cdot (-\varepsilon_K + \varepsilon_K) = 0$

2.2 Structural identity

$\{\hat{\gamma}^5, \hat{T}\} = 0$ on $H^2(D_{IV}^5)$.

Equivalently: $[\hat{\gamma}^5, \hat{T}] = -2 \hat{\gamma}^5 \hat{T}$ (or $2 \hat{T} \hat{\gamma}^5$, equivalent forms).

This is consistent with standard QFT Dirac algebra (γ^5 anti-commutes with γ^0 ; T-conjugation involves γ^0 structure).

3. Step 4 — $[\hat{\gamma}^5, \hat{C}]$

3.1 Computation

Apply $\hat{\gamma}^5 \circ \hat{C}$: $\hat{\gamma}^5 \hat{C} V_K = \hat{\gamma}^5 \cdot (\text{Pin}(2)\text{-}\hat{C}\text{-phase}) \cdot V_{(m_1, -m_2)} = (\text{Pin}(2)\text{-}\hat{C}\text{-phase}) \cdot \varepsilon_{\{\sigma K\}} \cdot V_{(m_1, -m_2)}$

Apply $\hat{C} \circ \hat{\gamma}^5$: $\hat{C} \hat{\gamma}^5 V_K = \hat{C} (\varepsilon_K \cdot V_K) = \varepsilon_K \cdot \hat{C} V_K = \varepsilon_K \cdot (\text{Pin}(2)\text{-}\hat{C}\text{-phase}) \cdot V_{(m_1, -m_2)}$

The key question (again): does $\hat{C}$ flip the $\text{Pin}(2)$ Z_2 grading? Per standard QFT: C takes $\psi_L \rightarrow \psi_R^c$ (left-handed particle to right-handed antiparticle), so C flips chirality on fermions.

$\varepsilon_{\{\sigma K\}} = -\varepsilon_K$ under $\hat{C}$ -action (chirality flips under C)

Anti-commutator: $\{\hat{\gamma}^5, \hat{C}\} V_K = \hat{\gamma}^5 \hat{C} V_K + \hat{C} \hat{\gamma}^5 V_K = (\text{Pin}(2)\text{-}\hat{C}\text{-phase}) \cdot V_{(m_1, -m_2)} \cdot (-\varepsilon_K + \varepsilon_K) = 0$

3.2 Structural identity

$\{\hat{\gamma}^5, \hat{C}\} = 0$ on $H^2(D_{IV}^5)$.

Equivalently: $[\hat{\gamma}^5, \hat{C}] = -2 \hat{\gamma}^5 \hat{C}$ (or $2 \hat{C} \hat{\gamma}^5$).

4. Step 5 — $[\hat{\gamma}^5, \hat{P}_{op}]$

4.1 Computation

Per Section 1.4: $\hat{P}_{op} = \hat{\gamma}^5 \circ \sigma$. Apply $\hat{\gamma}^5 \circ \hat{P}_{op}$: $\hat{\gamma}^5 \hat{P}_{op} V_K = \hat{\gamma}^5 \cdot \hat{\gamma}^5 \cdot \sigma V_K = (\hat{\gamma}^5)^2 \cdot \sigma V_K = \sigma V_K$ (since $(\hat{\gamma}^5)^2 = 1$)

Apply $\hat{P}_{op} \circ \hat{\gamma}^5$: $\hat{P}_{op} \hat{\gamma}^5 V_K = \hat{\gamma}^5 \cdot \sigma \cdot \hat{\gamma}^5 V_K = \hat{\gamma}^5 \cdot \sigma \cdot \varepsilon_K \cdot V_K = \hat{\gamma}^5 \cdot \varepsilon_K \cdot \sigma V_K = \varepsilon_K \cdot \hat{\gamma}^5 \cdot \sigma V_K$

Hmm — need to check whether σ commutes with $\hat{\gamma}^5$.

σ acts on the $\text{SO}(2)$ factor (Möbius involution: $z \rightarrow -\bar{z}$ in Hua coords). $\hat{\gamma}^5$ acts on the $\text{Pin}(2)$ Z_2 grading. These are conceptually independent operations (σ is on $\text{SO}(2)/U(1)$ part; $\hat{\gamma}^5$ is on $\text{Pin}(2)$ Z_2 lift).

Claim: σ commutes with $\hat{\gamma}^5$.

Substrate-mechanism: the Möbius involution preserves the $\text{Pin}(2)$ cover structure (it's a holomorphic-anti-holomorphic involution that doesn't act on the discrete $\mathbb{Z}_2$ grading). Therefore $[\sigma, \hat{\gamma}^5] = 0$.

Then: $\hat{P}_{\text{op}} \hat{\gamma}^5 V_K = \varepsilon_K \cdot \hat{\gamma}^5 \cdot \sigma V_K = \varepsilon_K \cdot \sigma \cdot \hat{\gamma}^5 V_K \cdot (1/\varepsilon_K) = \sigma \cdot \hat{\gamma}^5 V_K = \sigma \cdot \varepsilon_K V_K = \varepsilon_K \cdot \sigma V_K$

Wait, let me redo more carefully.

$\hat{\gamma}^5 \sigma V_K = \sigma \hat{\gamma}^5 V_K$ (since $[\sigma, \hat{\gamma}^5] = 0$) $= \sigma (\varepsilon_K V_K) = \varepsilon_K \cdot \sigma V_K$ (σ is linear so commutes with scalars)

So $\hat{\gamma}^5 \sigma V_K = \varepsilon_K \cdot \sigma V_K$.

Now: $\hat{P}_{\text{op}} \hat{\gamma}^5 V_K = \hat{\gamma}^5 \sigma \hat{\gamma}^5 V_K = \hat{\gamma}^5 \sigma \cdot \varepsilon_K V_K = \varepsilon_K \cdot \hat{\gamma}^5 \sigma V_K = \varepsilon_K \cdot \varepsilon_K \cdot \sigma V_K = \varepsilon_K^2 \cdot \sigma V_K = \sigma V_K$

$\hat{\gamma}^5 \hat{P}_{\text{op}} V_K = \hat{\gamma}^5 \hat{\gamma}^5 \sigma V_K = \sigma V_K$ (since $(\hat{\gamma}^5)^2 = 1$)

So both $\hat{\gamma}^5 \hat{P}_{\text{op}} V_K = \hat{P}_{\text{op}} \hat{\gamma}^5 V_K = \sigma V_K$.

Commutator: $[\hat{\gamma}^5, \hat{P}_{\text{op}}] = 0$.

4.2 Structural identity

$[\hat{\gamma}^5, \hat{P}_{\text{op}}] = 0$ on $H^2(D_{\text{IV}}^5)$.

$\hat{\gamma}^5$ COMMUTES with substrate-parity $\hat{P}_{\text{op}}$. This is because $\hat{P}_{\text{op}} = \hat{\gamma}^5 \circ \sigma$ already contains $\hat{\gamma}^5$ as a factor; commuting with $\hat{\gamma}^5$ just gives σ on both sides.

5. Composite structural reading — CPT theorem at substrate level

5.1 The pattern

Commutator	Result	Substrate-mechanism
$\{\hat{\gamma}^5, \hat{T}\} = 0$	Anti-commute	T flips $\text{Pin}(2)$ $\mathbb{Z}_2$ grading (S^1 rotation orientation reversal)
$\{\hat{\gamma}^5, \hat{C}\} = 0$	Anti-commute	C flips $\text{Pin}(2)$ $\mathbb{Z}_2$ grading (particle $\leftrightarrow$ antiparticle chirality flip)
$[\hat{\gamma}^5, \hat{P}_{\text{op}}] = 0$	Commute	$\hat{P}_{\text{op}}$ contains $\hat{\gamma}^5$ structurally ($\hat{P}_{\text{op}} = \hat{\gamma}^5 \circ \sigma$)

5.2 CPT-conjugate composite

The CPT-conjugate operator product: $\Theta_{\text{CPT}} := \hat{T} \cdot \hat{C} \cdot \hat{P}_{\text{op}}$ (or equivalently $\hat{P}_{\text{op}} \cdot \hat{C} \cdot \hat{T}$)

Compute commutator with $\hat{\gamma}^5$: $\hat{\gamma}^5 \Theta_{\text{CPT}} = \hat{\gamma}^5 \hat{T} \hat{C} \hat{P}_{\text{op}} = -\hat{T} \hat{\gamma}^5 \hat{C} \hat{P}_{\text{op}}$ (using $\{\hat{\gamma}^5, \hat{T}\} = 0$) $= -\hat{T} \cdot (-\hat{C} \hat{\gamma}^5) \cdot \hat{P}_{\text{op}}$ (using $\{\hat{\gamma}^5, \hat{C}\} = 0$) $= \hat{T} \hat{C} \hat{\gamma}^5 \hat{P}_{\text{op}} = \hat{T} \hat{C} \hat{P}_{\text{op}} \hat{\gamma}^5$ (using $[\hat{\gamma}^5, \hat{P}_{\text{op}}] = 0$) $= \Theta_{\text{CPT}} \hat{\gamma}^5$

So: $[\hat{\gamma}^5, \Theta_{\text{CPT}}] = 0$ — chirality COMMUTES with the CPT-conjugate operator.

This is the substrate-level CPT theorem: $\hat{\gamma}^5$ chirality is preserved by the combined CPT operation, even though individual $\hat{T}$ and $\hat{C}$ each flip chirality. The substrate's CPT structure is captured exactly by A_{sub} algebraic structure: the two minus signs from T and C cancel, and $\hat{P}_{\text{op}}$ preserves chirality.

5.3 CP-violation source structure

For CP only ($\hat{C} \cdot \hat{P}_{\text{op}}$): $\hat{\gamma}^5 \hat{C} \hat{P}_{\text{op}} = -\hat{C} \hat{\gamma}^5 \hat{P}_{\text{op}} = -\hat{C} \hat{P}_{\text{op}} \hat{\gamma}^5$

So $\{\hat{\gamma}^5, \hat{C} \hat{P}_{\text{op}}\} = 0$ — CP anti-commutes with $\hat{\gamma}^5$.

This is the substrate-algebraic source of observable CP-violation in chirality-distinguished processes. CP-violation observables in BST emerge from this $\{\hat{\gamma}^5, \text{CP}\} = 0$ anti-commutation; the standard Kobayashi-Maskawa CP-violation in weak interactions is the experimental projection of this substrate-level anti-commutation.

(Cal #27 STANDING reflexive trigger: this reading feels substrate-natural. Honest scope check — the anti-commutation IS the substrate-algebraic content; the CONNECTION to observable CP-violation requires explicit substrate-to-experimental projection map; multi-week INTERPRETIVE for that connection.)

6. Graph framing — edge structure from $\hat{\gamma}^5 \times$ discrete symmetries

Per A_{sub} v0.4 Section 3 graph framing:

Operator	K-type action	Edge type
$\hat{\gamma}^5$	Diagonal eigenvalue ± 1	Self-loop with sign
$\hat{T}$	$V \rightarrow \bar{V}$ (complex-conjugate rep) with m_2 sign-flip	Conjugate K-type edge
$\hat{C}$	$V \rightarrow V$ (with charge-flip phase)	Charge-flipped K-type edge
$**\hat{P}_{\text{op}}**$	$V \rightarrow V$ (with $\sigma + \hat{\gamma}^5$ phases)	Parity-flipped K-type edge

Edges from anti-commuting pairs ($\{\hat{\gamma}^5, \hat{T}\}$, $\{\hat{\gamma}^5, \hat{C}\}$): induce Z_2 -grading-flipping transitions between bosonic and fermionic sublattices.

Edges from commuting pair ($\{\hat{\gamma}^5, \hat{P}_{\text{op}}\}$): stay within same Z_2 -grading sublattice.

This refines A_sub v0.4 Section 2.2 partition: $\hat{P}_{\text{op}}$ preserves the boson/fermion distinction; $\hat{T}$ and $\hat{C}$ separately flip it (but CPT product preserves it).

7. Honest scope (Cal #27 STANDING)

What's STRUCTURALLY VERIFIED CANDIDATE in v0.1: - $\{\hat{\gamma}^5, \hat{T}\} = 0$ and $\{\hat{\gamma}^5, \hat{C}\} = 0$ anti-commutations (Sections 2-3): consistent with standard Dirac algebra; derivable from substrate structure given $\varepsilon_{\{\sigma K\}} = -\varepsilon_K$ for T-action and C-action - $[\hat{\gamma}^5, \hat{P}_{\text{op}}] = 0$ commutation (Section 4): direct from $\hat{P}_{\text{op}} = \hat{\gamma}^5 \circ \sigma$ structure - CPT composite Θ_{CPT} commutes with $\hat{\gamma}^5$ (Section 5.2): direct algebra from above three

What's MODEL-DEPENDENT in v0.1: - The claim $\varepsilon_{\{\sigma K\}} = -\varepsilon_K$ under T-action requires explicit substrate-mechanism for how $\hat{T}$ flips $\text{Pin}(2)$ Z_2 grading (Section 2.1). This is consistent with standard QFT but needs explicit substrate derivation for v0.2. Multi-week. - The claim $[\sigma, \hat{\gamma}^5] = 0$ (Möbius involution commutes with $\text{Pin}(2)$ Z_2 lift) is structurally natural but needs explicit verification for v0.2.

What's INTERPRETIVE in v0.1: - Section 5.3 substrate-level $\{\hat{\gamma}^5, \text{CP}\} = 0$ as source of observable CP-violation: claim consistent with algebra but requires explicit projection-map analysis (multi-week) - The substrate-mechanism reading for T's grading-flip via S^1 orientation reversal: substrate-natural but unverified

Cal #27 STANDING reflexive trigger count: 2 triggers (substrate-CPT structure + CP-violation source). Both honest-scope checked — algebraic content is RATIFIED-standard; experimental-projection claims are INTERPRETIVE pending explicit projection-map work.

Forward-derivation discipline: results derived from A_sub algebra without back-fitting to target observables. The pattern $\{\hat{\gamma}^5, \hat{T}\} = \{\hat{\gamma}^5, \hat{C}\} = 0 + [\hat{\gamma}^5, \hat{P}_{\text{op}}] = 0 + [\hat{\gamma}^5, \text{CPT}] = 0$ mirrors standard QFT CPT theorem — this is structural concord, not back-fit.

8. Cumulative A_sub commutator status

#	Commutator	Result	Status
1	$[\hat{Q}, \hat{P}_{\text{op}}]$	$\{\hat{Q}, \hat{P}_{\text{op}}\} = 0$ (anti-commute)	STRUCTURALLY VERIFIED CANDIDATE (step 1 v0.1)
2	$[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$	$-(2 \hat{Q} + N_{\text{c}} - 1) \cdot \hat{T}_{\text{tick}}$	STRUCTURALLY VERIFIED CANDIDATE (step 2 v0.1, model- dependent)

#	Commutator	Result	Status
3	$[\hat{\gamma}^5, \hat{T}]$	$\{\hat{\gamma}^5, \hat{T}\} = 0$ (anti-commute)	STRUCTURALLY VERIFIED CANDIDATE (step 3 v0.1)
4	$[\hat{\gamma}^5, \hat{C}]$	$\{\hat{\gamma}^5, \hat{C}\} = 0$ (anti-commute)	STRUCTURALLY VERIFIED CANDIDATE (step 4 v0.1)
5	$[\hat{\gamma}^5, \hat{P}_{op}]$	$[\hat{\gamma}^5, \hat{P}_{op}] = 0$ (commute; already- contained)	STRUCTURALLY VERIFIED CANDIDATE (step 5 v0.1)
6	$[\hat{B}, \hat{T}_{tick}]$	TBD — Track DC load-bearing for Bell 1/8 mechanism	PENDING
7	$[\hat{L}_i, \hat{\gamma}^5]$	TBD — spin-orbit coupling structure	PENDING
8	$[\hat{B}, \hat{Q}]$	TBD — Bell-test charge sensitivity	PENDING
9	$[\hat{C}_3, \hat{L}_3]$	TBD — electroweak gauge non- commutativity	PENDING

5 of 9 A_{sub} commutators closed. 4 remain.

9. Recommended next pull

$[\hat{B}, \hat{T}_{tick}]$ — Track DC load-bearing for Bell 1/8 dynamical mechanism. Requires careful $\hat{T}_{tick}$ model handling (Section 1.1 step 2 noted simplest-model dependency). Next session work.

Alternative: $[\hat{L}_i, \hat{\gamma}^5]$ spin-orbit coupling. $SO(5)$ angular momentum $\times$ $Pin(2)$ Z_2 chirality. Standard QM substantive piece — explains atomic spectroscopy via substrate algebra. Less load-bearing for Track DC but valuable for Vol 3 nuclear/atomic + Vol 5 QM cross-references.

10. Coordination

Cal: cold-read of Sections 2-5 algebra (mechanical) + tier-discipline check on Section 5.2-5.3 CPT-structure + CP-violation source interpretive content. Type C level-

crossing per Cal #122 if CP-violation source claim supports Strong-Uniqueness extension (likely not — substrate-derived from existing T2471 + T2433 + T2434 + T2472).

Keeper: K-audit chain entry for steps 3-5 (batch) $\hat{\gamma}^5 \times$ discrete-symmetry pattern.

Elie: K52a Session 7+ may inform substrate-CPT structure; Toy 3531+ population principle benefits from $\hat{\gamma}^5 \times$ discrete-symmetry constraints on K-type accessibility.

Grace: catalog entries for $\{\hat{\gamma}^5, \hat{T}\} = \{\hat{\gamma}^5, \hat{C}\} = 0 + [\hat{\gamma}^5, \hat{P}_{op}] = 0 + [\hat{\gamma}^5, CPT] = 0$ substrate-CPT structure.

— Lyra, A_sub commutator closures steps 3-5 batched v0.1 filed Tuesday 2026-05-26 ~08:55 EDT per Task #322 v0.4 Section 3.3 work plan. STRUCTURALLY VERIFIED CANDIDATE pending Cal cold-read of Sections 2-5 algebra. Substantive composite finding: $\hat{\gamma}^5$ anti-commutes with $\hat{T}$ and $\hat{C}$ individually but commutes with CPT composite $\Theta_{CPT} = \hat{T} \cdot \hat{C} \cdot \hat{P}_{op}$ — substrate-level CPT theorem at the algebraic level. CP-only anti-commutes with $\hat{\gamma}^5$ — substrate-algebraic source of observable CP-violation.